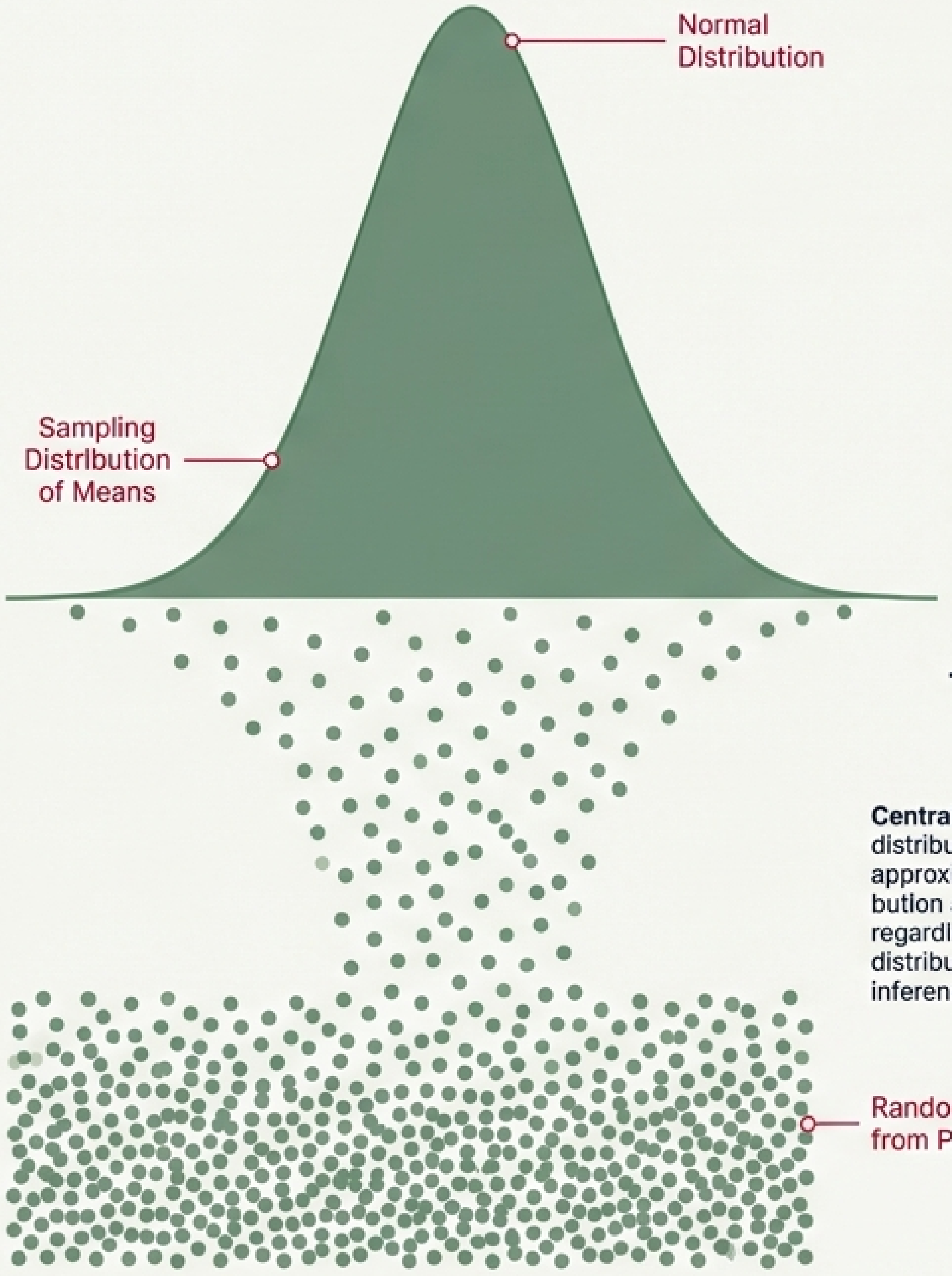
Editorial Science

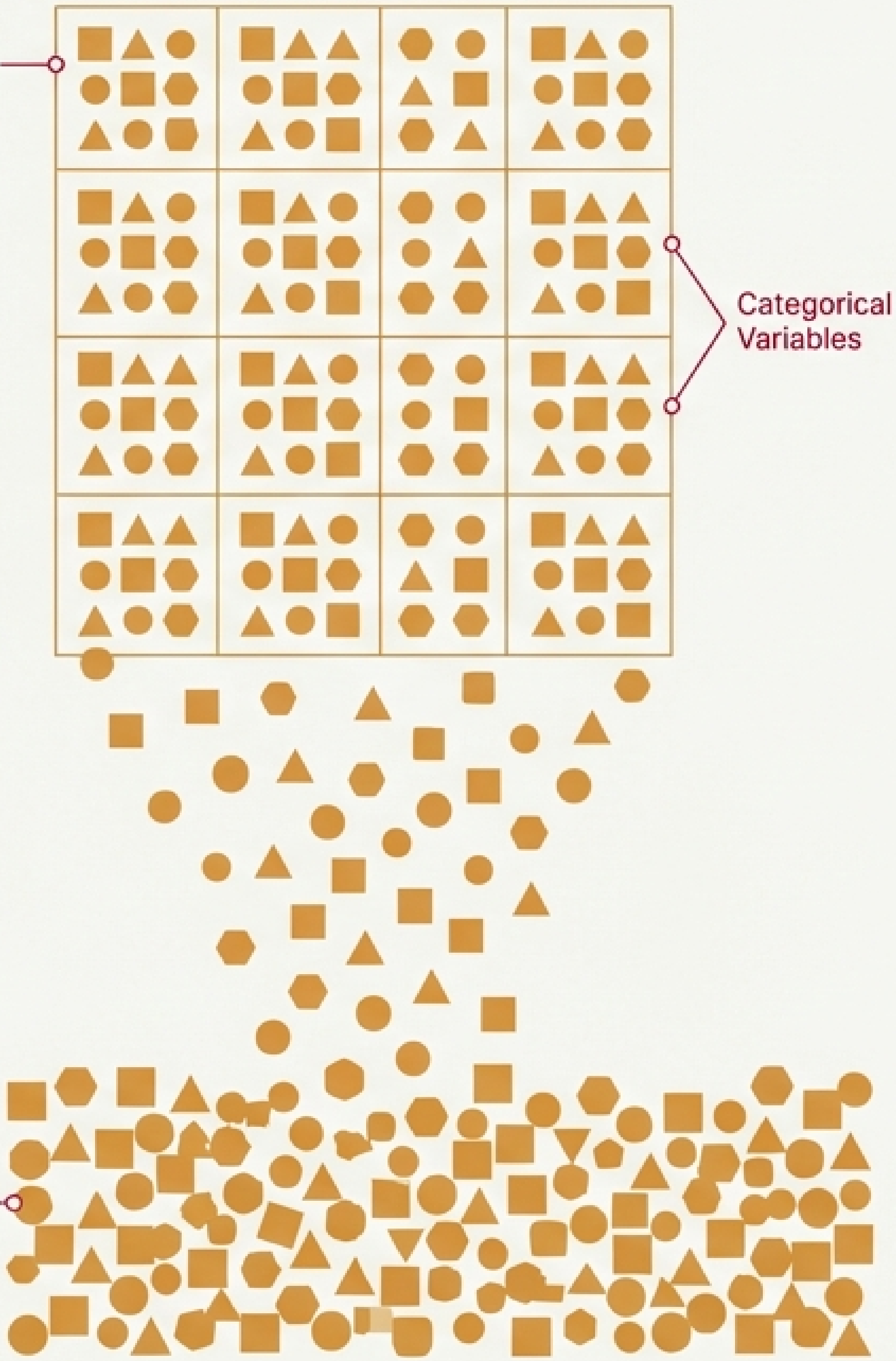



# Statistical Inference Essentials

## Mastering the Central Limit Theorem & Chi-Square Tests

**Central Limit Theorem:** The distribution of sample means approximates a normal distribution as sample size increases, regardless of the population's distribution, enabling reliable inference on continuous data.

**Chi-Square Tests:** Assess the association or independence between categorical variables by comparing observed frequencies to expected frequencies, identifying significant patterns in categorical data.

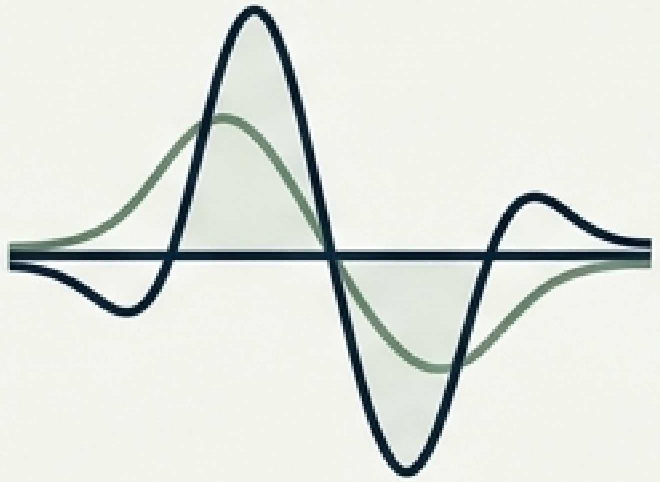



*From Randomness to Reliability*

# Making Sense of Randomness

Data is naturally messy. To find the signal in the noise, we rely on two pillars of statistical inference.

## Continuous Data



Tool: Central Limit Theorem (CLT)

Application: Making population inferences from samples.

## Categorical Data

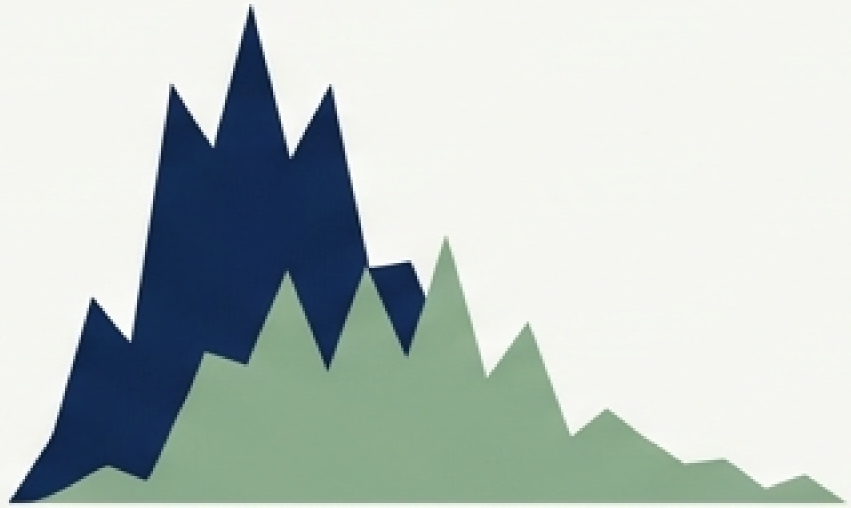
Tool: Chi-Square Tests

Application: Determining associations and goodness of fit.

**Key Insight:**
These methodologies form the backbone of A/B testing, predictive modelling, and hypothesis testing.

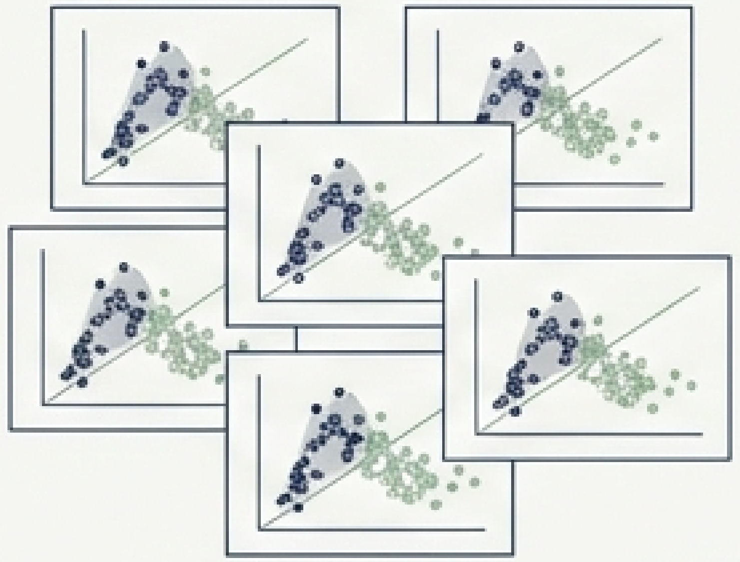
# The Power of the Sample

Any Population Distribution
(Skewed/Weird)

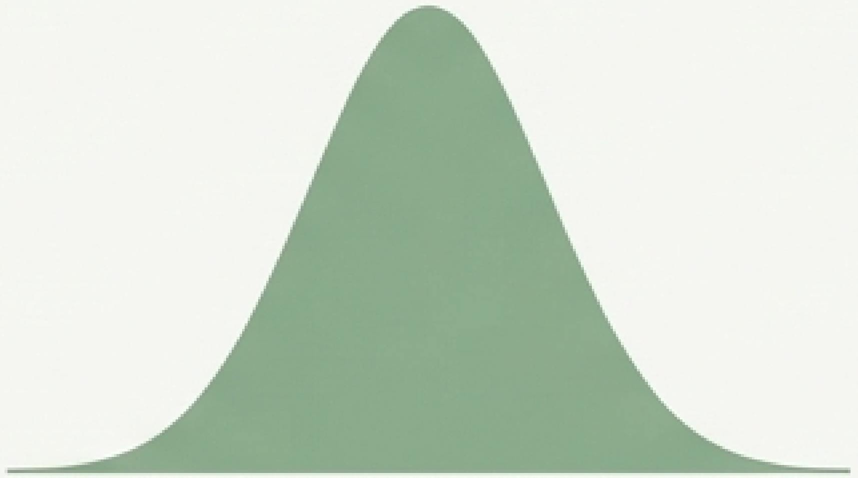


Sampling
Process

**Magic Threshold: $n \geq 30$**
Inter Tight



Sampling Distribution
of the Mean



The Central Limit Theorem (CLT) states that when we take random samples of a sufficiently large size ($n \geq 30$) from any population distribution, the sampling distribution of the sample mean will approximate a normal distribution.

$$\mathrm{SE} = \frac{\sigma}{\sqrt{n}}$$

# Why CLT Matters in Data Science

## The Reality: Student Heights

## The Inference


### Inference

Estimate population parameters from small samples.

### Tools

Unlocks Z-scores, Confidence Intervals, and T-tests on non-normal data.

### Testing

Essential for Hypothesis Testing and A/B Testing conversion rates.

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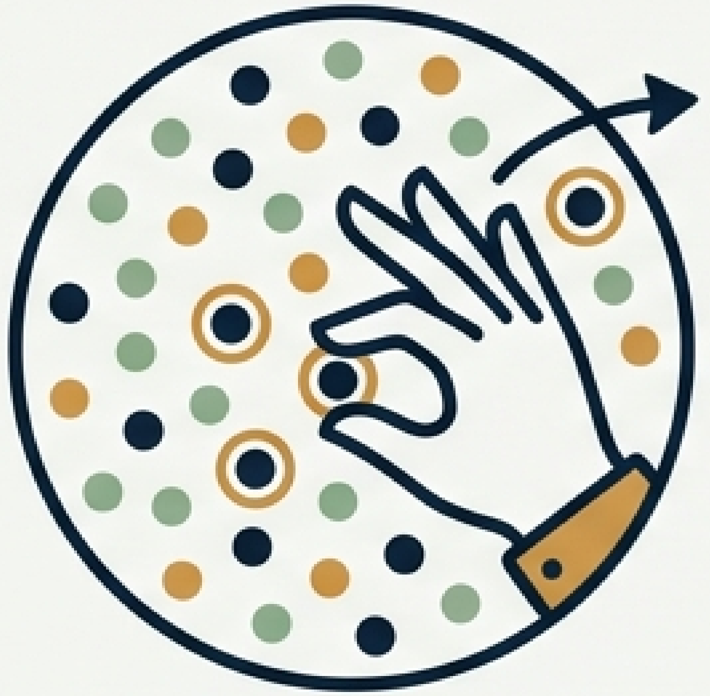
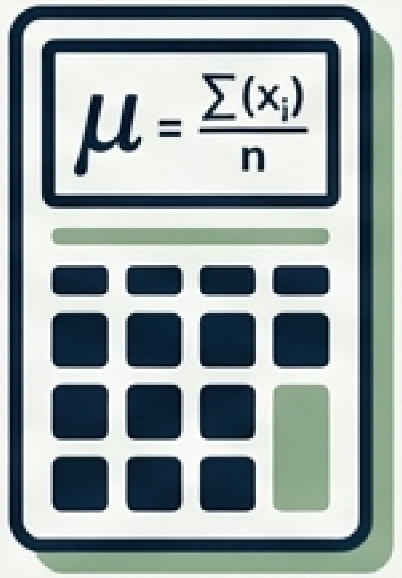
# The Sampling Process Step-by-Step

Stage 1
**Pick Samples**

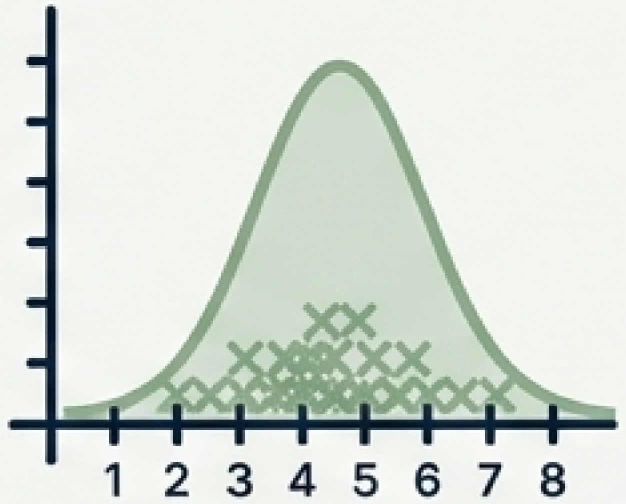


Stage 2
**Calculate Means**




Stage 3
**Plot Distribution**


Stage 4
**Observe Shape**

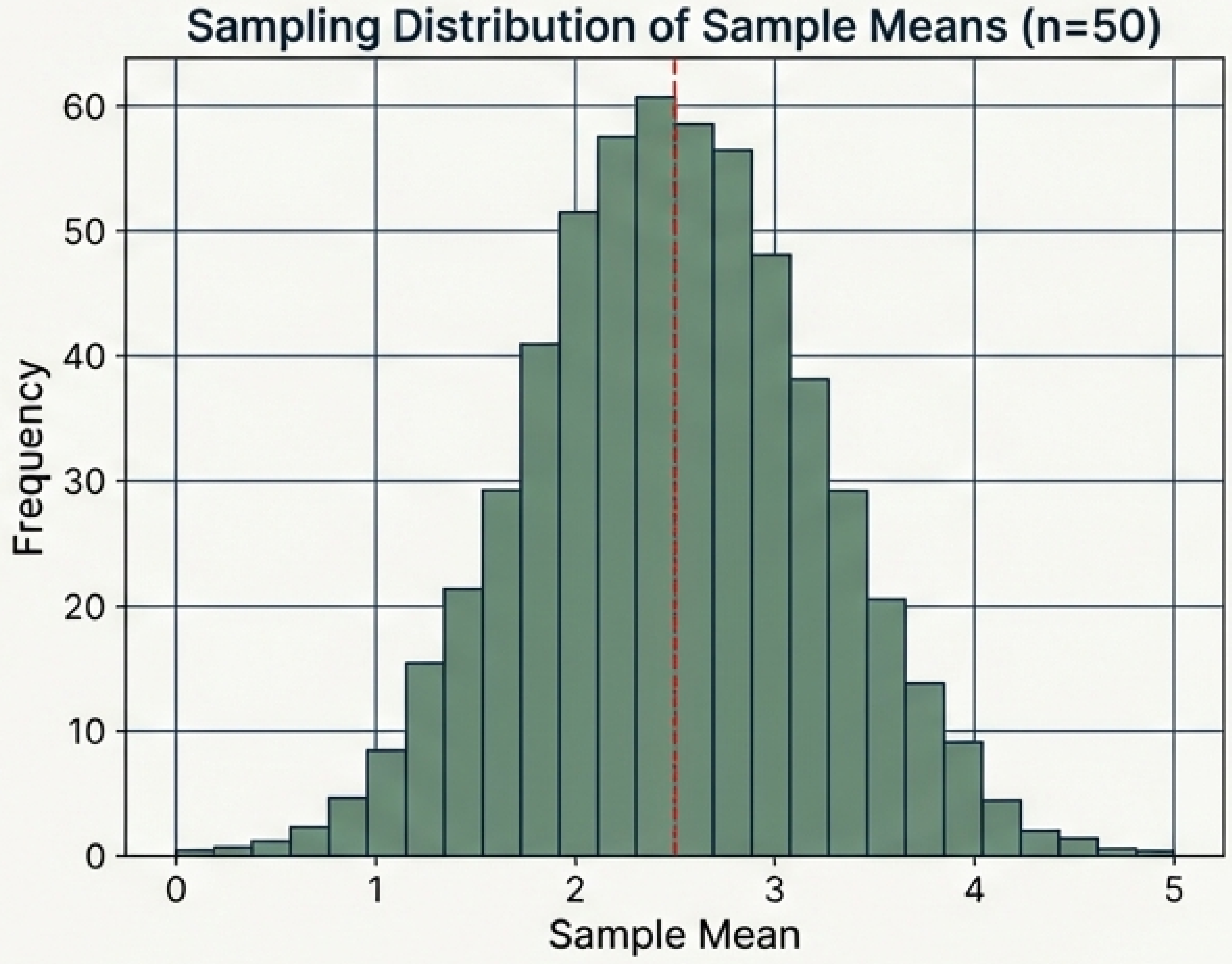



## Note on Prediction

Predictive modelling assumes error terms are normally distributed, a property often justified by CLT.

# Simulating CLT in Python

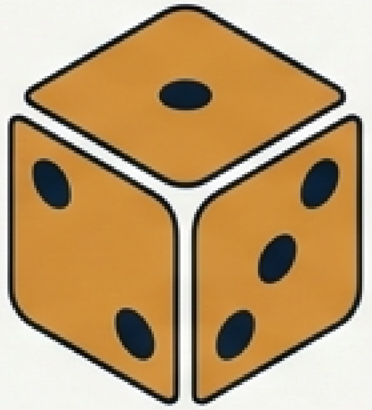
```
population = np.random.exponential(scale=2,
size=100000)

sample_means = []

for _ in range(1000):
    sample = np.random.choice(population,
    size=50)
    sample_means.append(np.mean(sample))

plt.hist(sample_means, bins=30)
```



# Analysing Categorical Data

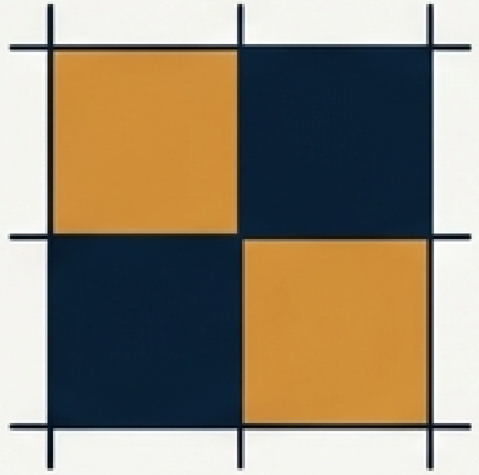
**When data is not numerical (like height) but categorical (Yes/No, Red/Blue), we cannot use CLT. We use Chi-Square.**

## Goodness of Fit



Tests if observed frequencies match expected frequencies (1 variable)

## Test of Independence



Tests if two variables are related (2 variables)

# Type I: Goodness of Fit Test

## The “Fairness” Test

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Observed Frequency
Crimson Pro
(What actually happened)

Expected Frequency
(Theory/Hypothesis)

The test sums the squared differences. If Observed and Expected are close, the result is small (Good Fit). If distinct, the result is large (Bad Fit).

# Example: Is the Die Fair?

| Face | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Observed - 60 rolls | 10 | 8 | 12 | 11 | 9 | 10 |
| Expected - Theory | 10 | 10 | 10 | 10 | 10 | 10 |

## The Calculation

$$\chi^2 = \frac{(10-10)^2}{10} + \frac{(8-10)^2}{10} + \cdots + \frac{(10-10)^2}{10}$$
$$= 0 + 0.4 + 0.4 + 0.1 + 0.1 + 0 = 1.0$$

## Chi-Square = 1.0

**Decision Logic:**

Critical Value (df=5, α=0.05) = 11.07

1.0 < 11.07 -> Fail to Reject Null Hypothesis.

**Conclusion:** The die appears fair.

# Goodness of Fit in Python

## Python Implementation

```
from scipy.stats import chisquare

observed = [10, 8, 12, 11, 9, 10]
expected = [10] * 6

chi_stat, p_value =
chisquare(f_obs=observed,
f_exp=expected)
```

## Test Results

```
>>> Chi-square statistic: 1.0
>>> p-value: 0.96 (High)

Conclusion: No significant
difference.
```

# Type II: Test of Independence

## The “Relationship” Test

Column Variable

Row Variable

| | | | T |
|---|---|---|---|
| | A1 | A2 | AT |
| | B1 | B2 | BT |
| T | AT | CT | GT |

Purpose: Tests if Variable A influences Variable B.

## The Math

The test statistic is the same, but the Expected Frequency calculation changes.

$$E_{ij} = \frac{\text{Row Total} \times \text{Col Total}}{\text{Grand Total}}$$

# Example: Gender vs. Drink Preference

Does gender influence the choice between Tea or Coffee?

| | Tea | Coffee | | |
|---|---|---|---|---|
| Male | 30 | 20 | 50 | Row Total |
| Female | 10 | 40 | 50 | Row Total |
| | 40 | 60 | 100 | Grand Total |
| | | Column Total | | |

# Calculating Independence

## Step 1: Expected Frequencies

$$E_{\text{Male,Tea}} = \frac{50 \times 40}{100} = 20$$

## Step 2: Chi-Square Statistic

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}} = \frac{(30-20)^2}{20} + \frac{(20-30)^2}{30} + \frac{(10-20)^2}{20} + \frac{(40-30)^2}{30}$$
$$= (100/20) + (100/30) + (100/20) + (100/30) = 5 + 3.33 + 5 + 3.33 = 16.66$$

**Result: Chi-Square = 16.66**

## Decision

Critical Value (df=1, α=0.05) = 3.84

16.66 > 3.84 → Reject Null Hypothesis.

**Conclusion:** Strong association exists between gender and preference.

# Test of Independence in Python

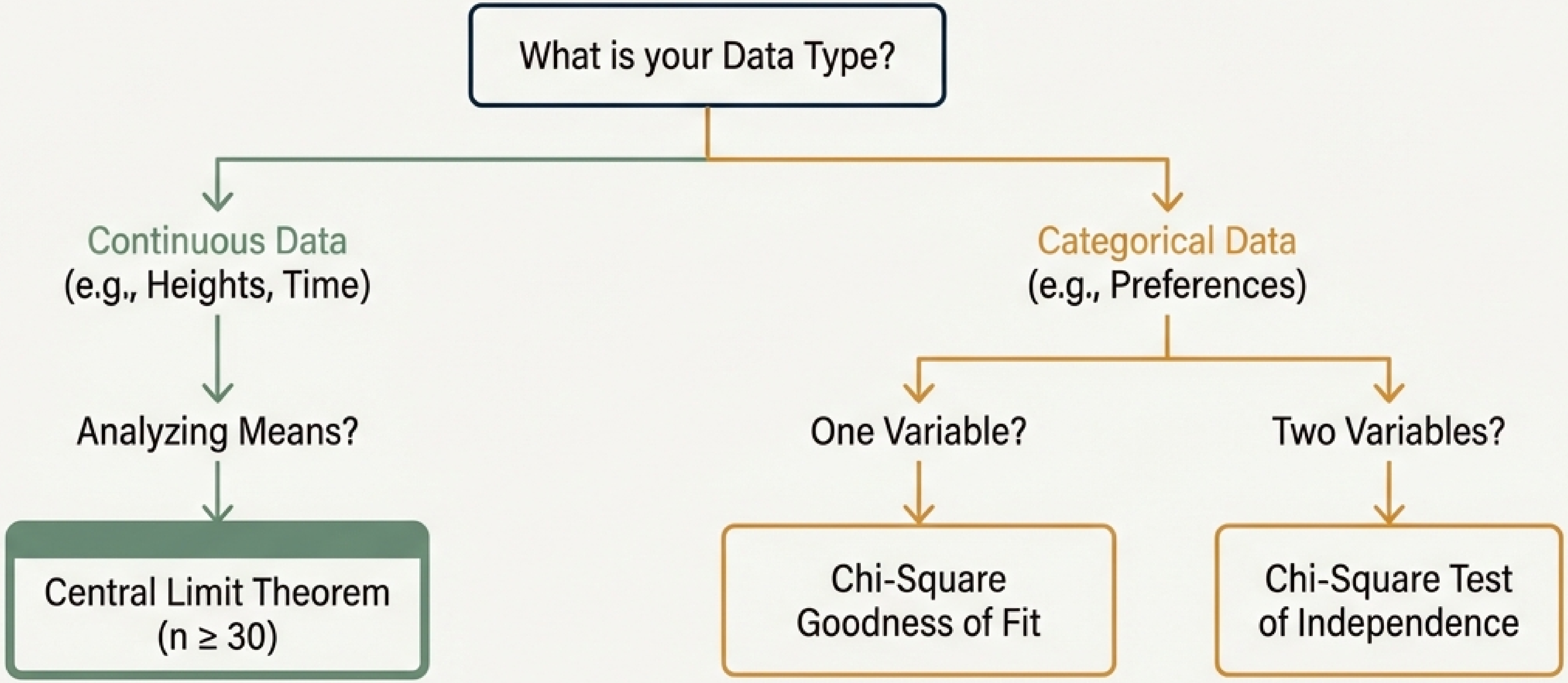
```
import pandas as pd
from scipy.stats import
chi2_contingency

data = [[30, 20], [10, 40]]
df = pd.DataFrame(data, columns=['Tea',
'Coffee'], index=['Male', 'Female'])

chi2, p, dof, expected =
chi2_contingency(df)
```

>>> Chi-square statistic: **16.66**

>>> p-value: 0.00004 (Significant)

Conclusion: **Reject Null Hypothesis.**

# The Inference Decision Matrix




These tests provide the mathematical confidence required to move from data observation to strategic decision-making